BoT-IoT MAQT Training Setups

This note summarizes how lawun chose MAQT training settings on FROZEN BoT-IoT, and how those runs compare to arjun's reported CE-head test accuracy.

Latest (full known train, cosine start 0.01): about 68% CE-head accuracy on the full known test with no reject mask — up from the previous coreset pick (3000 / class, same LR recipe, about 65%). See section 7.

Unless stated otherwise, confusion matrices below are Algo-3 accepted-known test results: known-class test rows with nonconformity $s(x) \leq q$ (Global conformal). Rejected known-test rows are excluded from accuracy and from the matrix.

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1. Baseline: 4-class, 3000 / class, fixed LR 0.05

Setup: four known classes (DDoS, DoS, Normal, Reconnaissance); DPP coreset capped at 3000 per class; fixed learning rate 0.05.

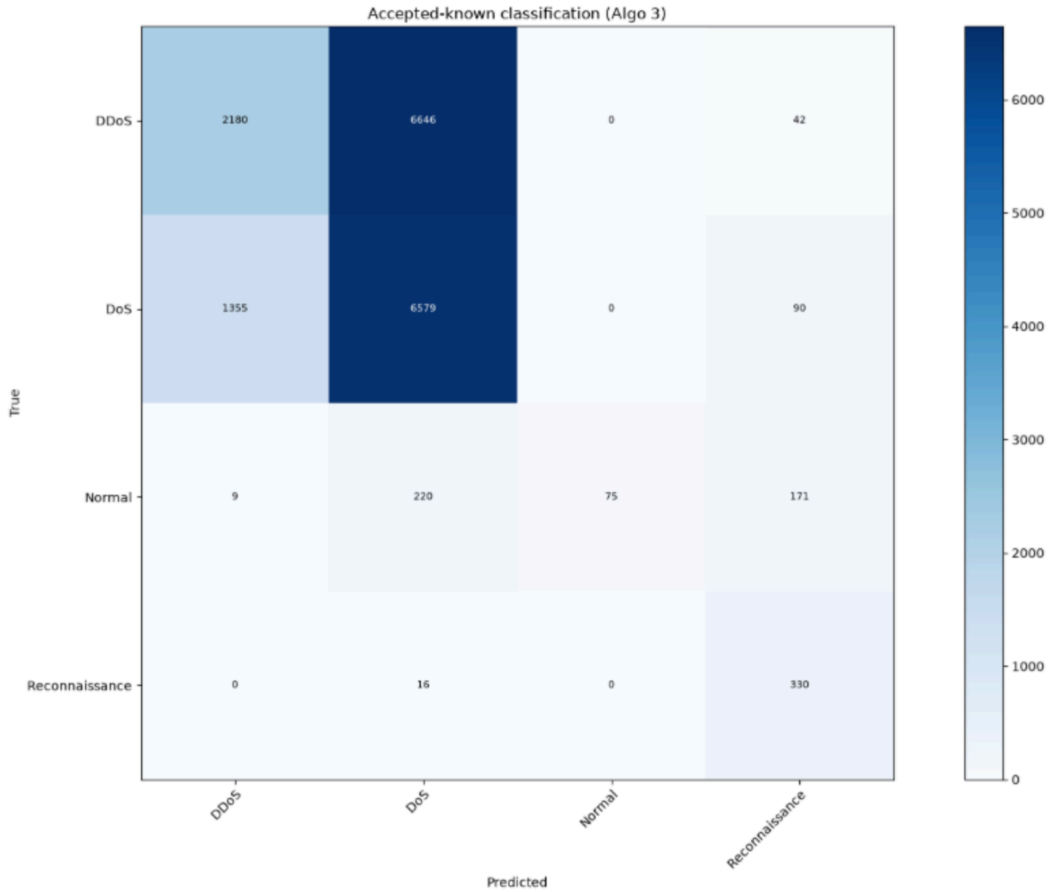
Observation: overall accepted accuracy is modest (~52%). Normal is poorly recovered (high precision, very low recall). DDoS and DoS are heavily confused.

```
[test] mean certified radius (accepted): 0.0125
[test] reject rate (known): 0.0472 | acc (accepted): 0.5174

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.62      0.25      0.35      8868
           DoS       0.49      0.82      0.61      8024
        Normal       1.00      0.16      0.27       475
Reconnaissance       0.52      0.95      0.67       346

      accuracy                           0.52     17713
     macro avg       0.66      0.54      0.48     17713
  weighted avg       0.57      0.52      0.47     17713
```



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2. Ablation: drop DoS (3-class), same budget

Hypothesis: majority DoS / DDoS traffic might be starving Normal.

Setup: train without DoS (DDoS + Normal + Reconnaissance only); still 3000 / class, LR 0.05.

Observation: Normal does not improve (accepted Normal recall stays at zero in this run). High headline accuracy is driven by DDoS dominance, not better Normal geometry.

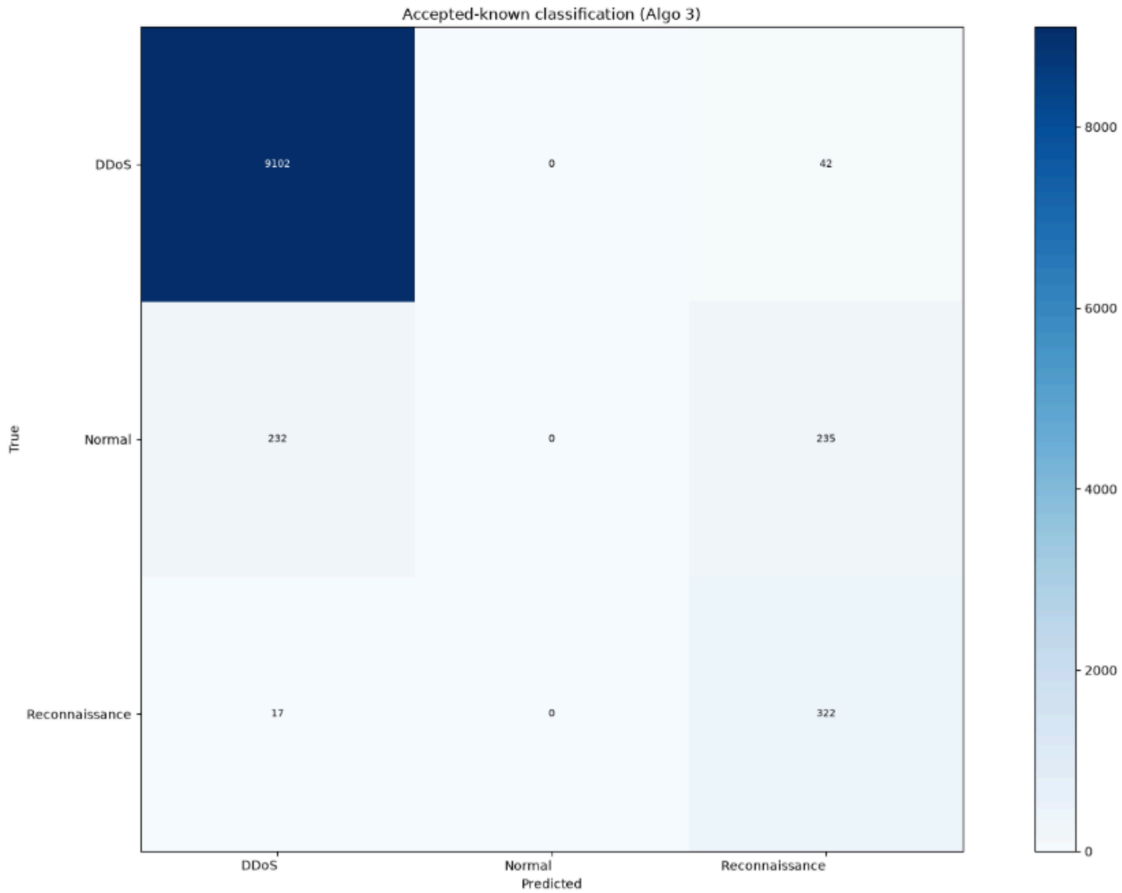
Conclusion: DoS presence is not the main cause of poor Normal prediction $\rightarrow$ return to 4-class.

```
[test] mean certified radius (accepted): 0.0710
[test] reject rate (known): 0.0484 | acc (accepted): 0.9471

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.97      1.00      0.98      9144
        Normal       0.00      0.00      0.00       467
Reconnaissance       0.54      0.95      0.69       339

      accuracy                           0.95      9950
     macro avg       0.50      0.65      0.56      9950
  weighted avg       0.91      0.95      0.93      9950
```



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3. Scale data: 4-class, 20000 / class, LR 0.05

Setup: larger DPP coreset (20000 per class), same fixed LR 0.05.

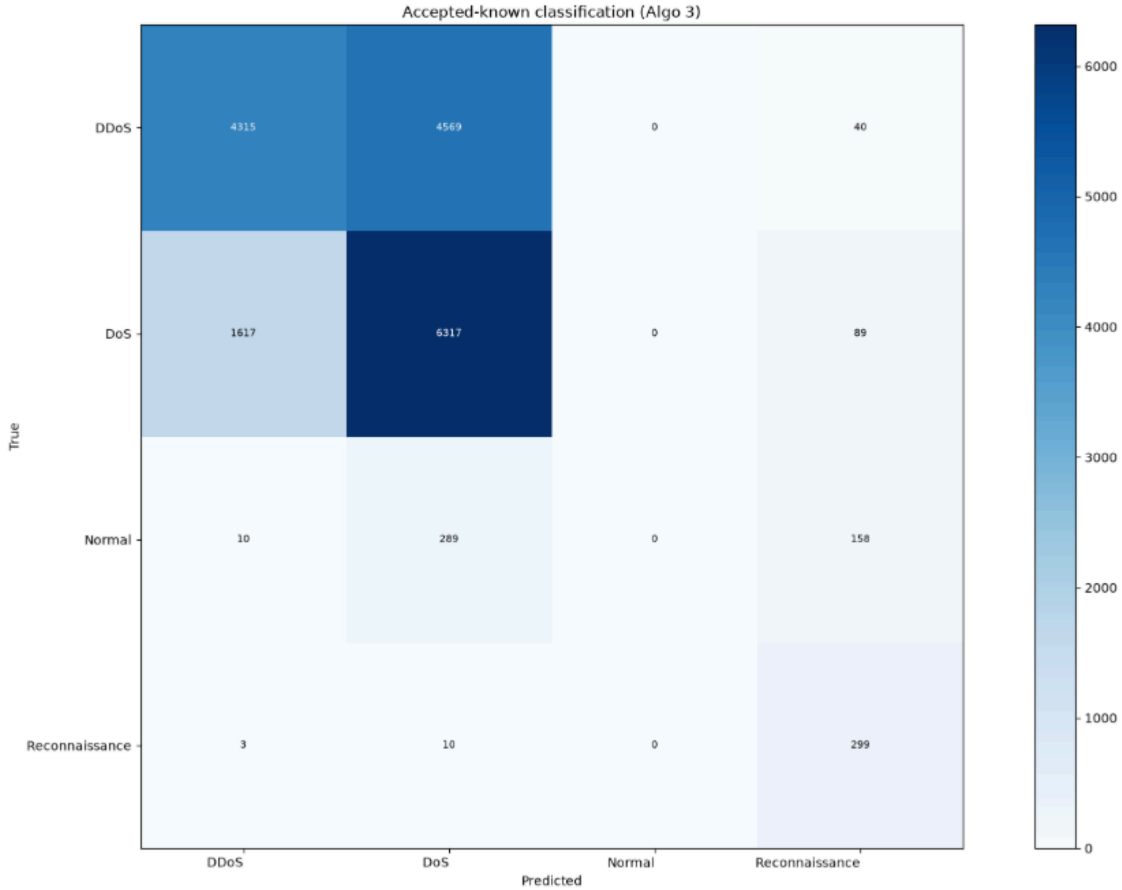
Observation: accepted accuracy rises (~62% vs ~52% at 3000). DDoS ↔ DoS confusion remains large; Normal stays weak (near-zero correct in this matrix). More data helps overall accuracy but does not fix class separation.

```
[test] mean certified radius (accepted): 0.0082
[test] reject rate (known): 0.0471 | acc (accepted): 0.6170

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.73      0.48      0.58      8924
           DoS       0.56      0.79      0.66      8023
        Normal       0.00      0.00      0.00       457
Reconnaissance       0.51      0.96      0.67       312

      accuracy                           0.62     17716
     macro avg       0.45      0.56      0.48     17716
  weighted avg       0.63      0.62      0.60     17716
```



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4. Smaller fixed LR, moderate size: 10000 / class, LR 0.0005

Setup: 10000 per class, fixed LR 0.0005.

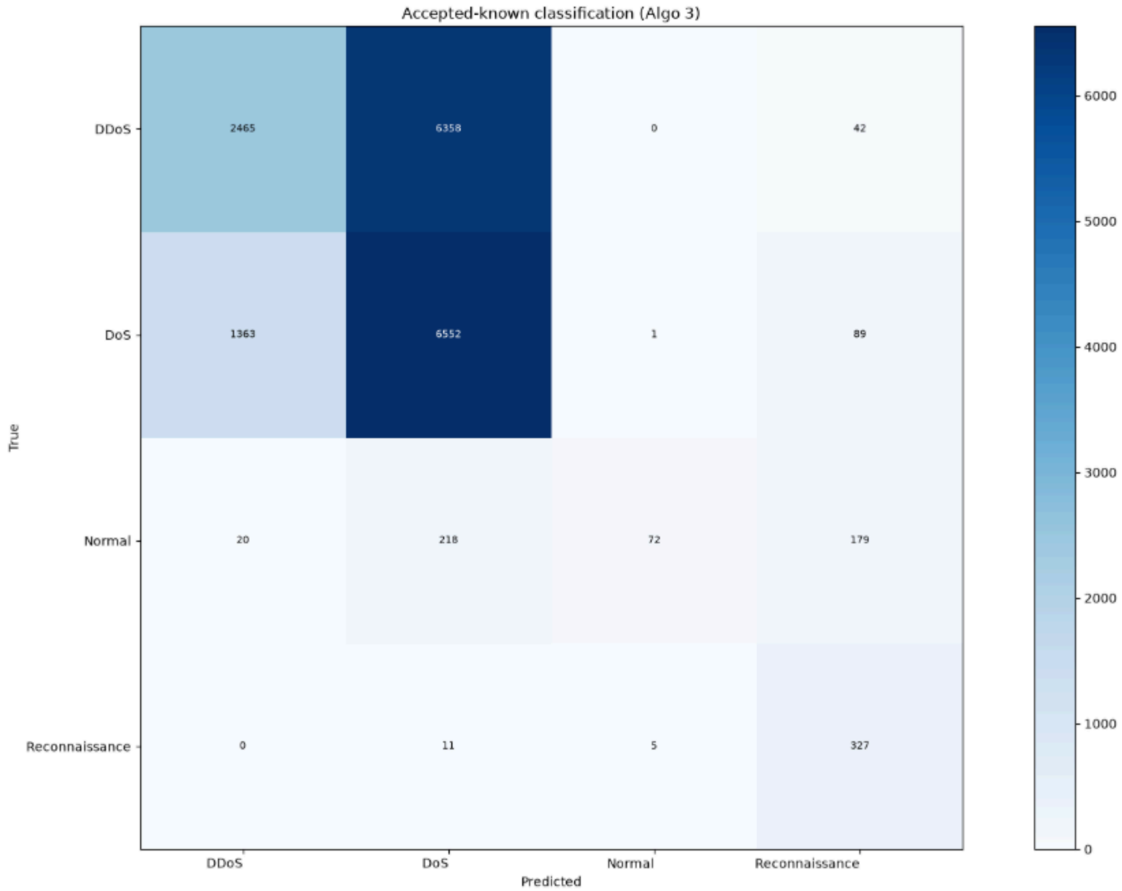
Observation: accepted accuracy (~53%) is close to the 3000 / LR 0.05 baseline (only a small bump, on the order of about one percentage point in this comparison). DDoS ↔ DoS and weak Normal persist. Extra samples + tiny fixed LR do not justify the cost for this pipeline.

```
[test] mean certified radius (accepted): 0.0132
[test] reject rate (known): 0.0478 | acc (accepted): 0.5319

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.64      0.28      0.39      8865
           DoS       0.50      0.82      0.62      8005
        Normal       0.92      0.15      0.25       489
Reconnaissance       0.51      0.95      0.67       343

      accuracy                           0.53     17702
     macro avg       0.64      0.55      0.48     17702
  weighted avg       0.58      0.53      0.49     17702
```



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5. Back to 3000 / class with cosine LR schedule

Given diminishing returns from larger coresets under fixed LR, lawun returns to 3000 per class and turns on automatic LR decay (`use_lr_schedule=True`, cosine annealing from the starting LR toward a small floor).

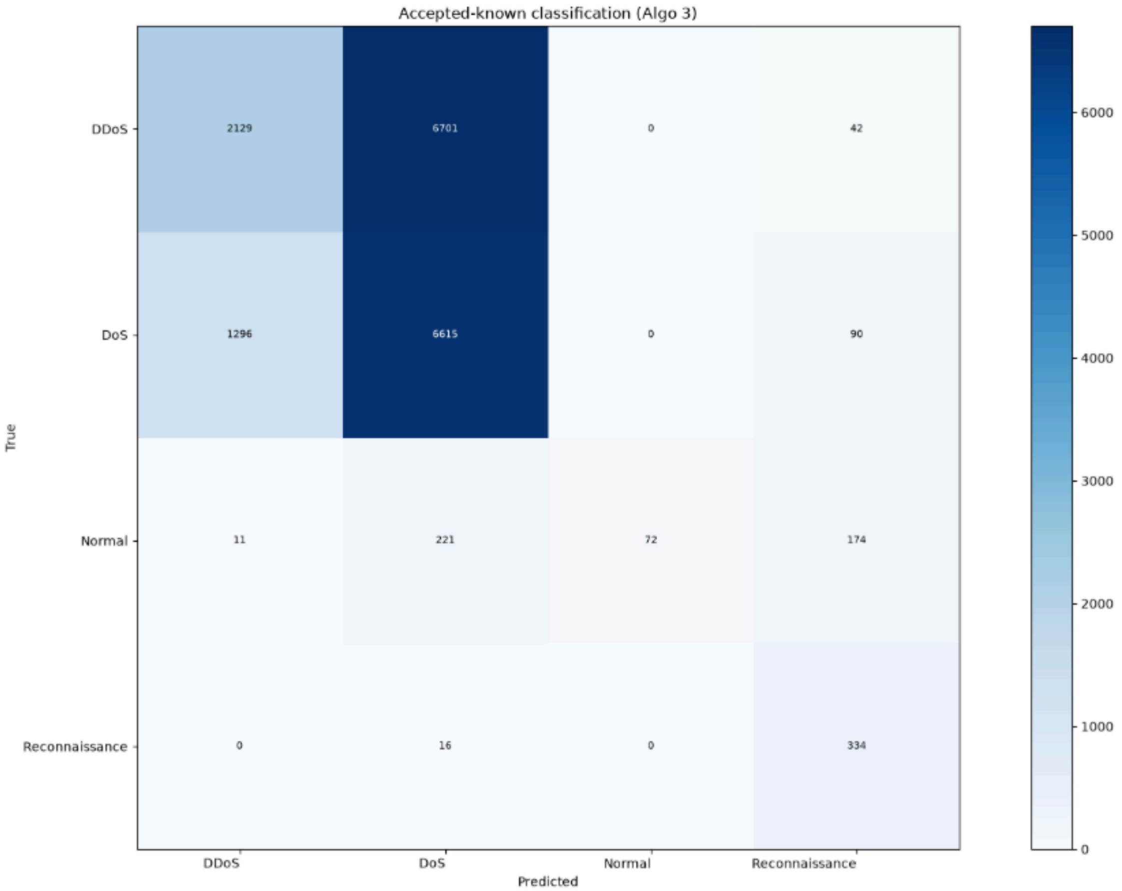
5a. Start LR 0.03

```
[test] mean certified radius (accepted): 0.0136
[test] reject rate (known): 0.0479 | acc (accepted): 0.5169

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.62      0.24      0.35      8872
           DoS       0.49      0.83      0.61      8001
        Normal       1.00      0.15      0.26       478
Reconnaissance       0.52      0.95      0.67       350

      accuracy                           0.52     17701
     macro avg       0.66      0.54      0.47     17701
  weighted avg       0.57      0.52      0.47     17701
```



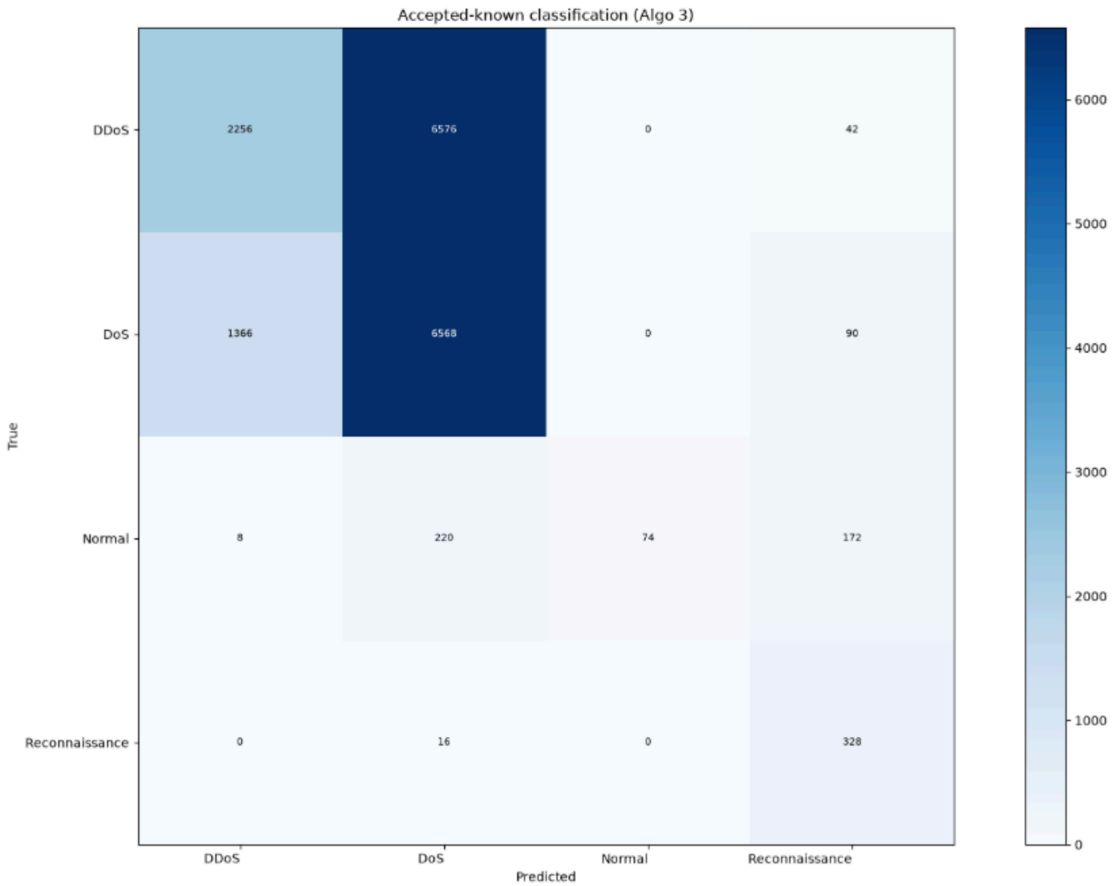
5b. Start LR 0.01 (chosen)

```
[test] mean certified radius (accepted): 0.0120
[test] reject rate (known): 0.0471 | acc (accepted): 0.5208

[test] classification report (accepted only):
                precision    recall  f1-score   support

          DDoS       0.62      0.25      0.36      8874
           DoS       0.49      0.82      0.61      8024
        Normal       1.00      0.16      0.27       474
Reconnaissance       0.52      0.95      0.67       344

      accuracy                           0.52     17716
     macro avg       0.66      0.55      0.48     17716
  weighted avg       0.57      0.52      0.48     17716
```



Observation: accepted metrics for start 0.03 vs 0.01 are similar (~52%). Training with start 0.01 is judged smoother on the loss/LR curves, so the coreset-era working choice was:

Knob	Choice
Classes	4 (DDoS, DoS, Normal, Reconnaissance)
Train coreset	3000 / class (DPP)
LR	cosine schedule, start 0.01
Notebooks	`4class-3000each-lr0.01auto-maqt` → `4class-3000each-lr0.01auto-algo3` (+ cached variants)

This was later superseded by full-train under the same LR recipe (section 7).

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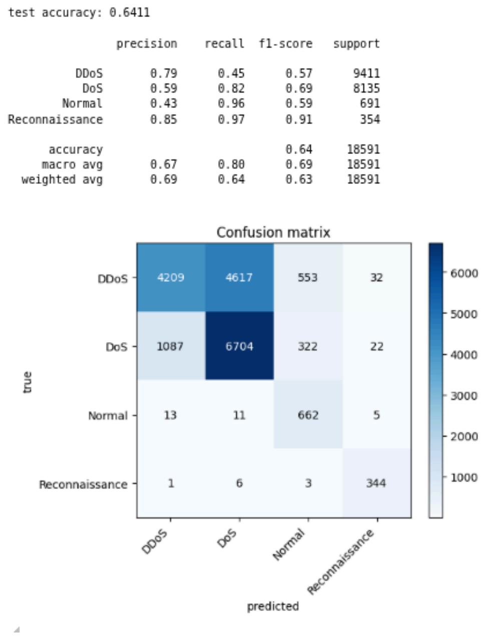
6. Fair compare to arjun (full known test, no reject mask)

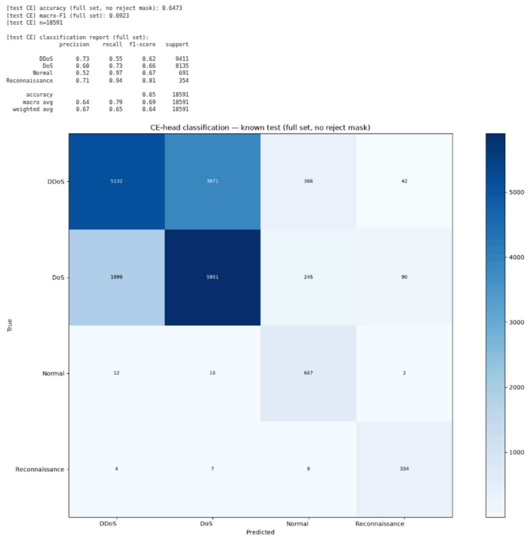
Algo-3 accepted-only accuracy is not comparable to a CE-head score on every known-test row.

- arjun (`ablation-test-bot-iot`): CE-head `predict_labels` on full known test $\to$ about 64% accuracy ($n = 18591$).
- lawun best MAQT (4-class, 3000 / class, cosine start 0.01): CE-head on the same full known test, no conformal mask $\to$ about 65% accuracy ($n = 18591$).

arjun (~64%, full known test, CE, no reject)



lawun (~65%, full known test, CE, no reject)



Under that matched protocol, the 3000 / class run was on par with (slightly above) arjun's reported CE full-test accuracy. Remaining open issue in those coresets: DDoS vs DoS separation; conformal accepted-only CMs still showed weak Normal recall when rejects are stripped out.

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7. Latest: full train (~148k), cosine start 0.01 → ~68% CE full-test

Motivation: the section 5 and 6 working choice (3000 / class + cosine 0.01, ~65% CE full-test) left headroom if the model saw all FROZEN known-train rows instead of a DPP coreset.

Setup: 4-class; `SUBSET=False` (full known train, $\sim$148k); cosine LR schedule, start 0.01; batch 64; same FROZEN features / circuit recipe as `final-bot-iot-maqt` / `4class-all-lr0.01auto-lambda2-maqt`.

Protocol (matched to section 6): CE-head on full known test, no conformal reject mask ($n = 18591$).

Metric	3000 / class, cosine 0.01 (section 6)	Full train, cosine 0.01 (this)
Accuracy	~65%	~68.2%
Macro-F1	—	~0.73
n	18591	18591

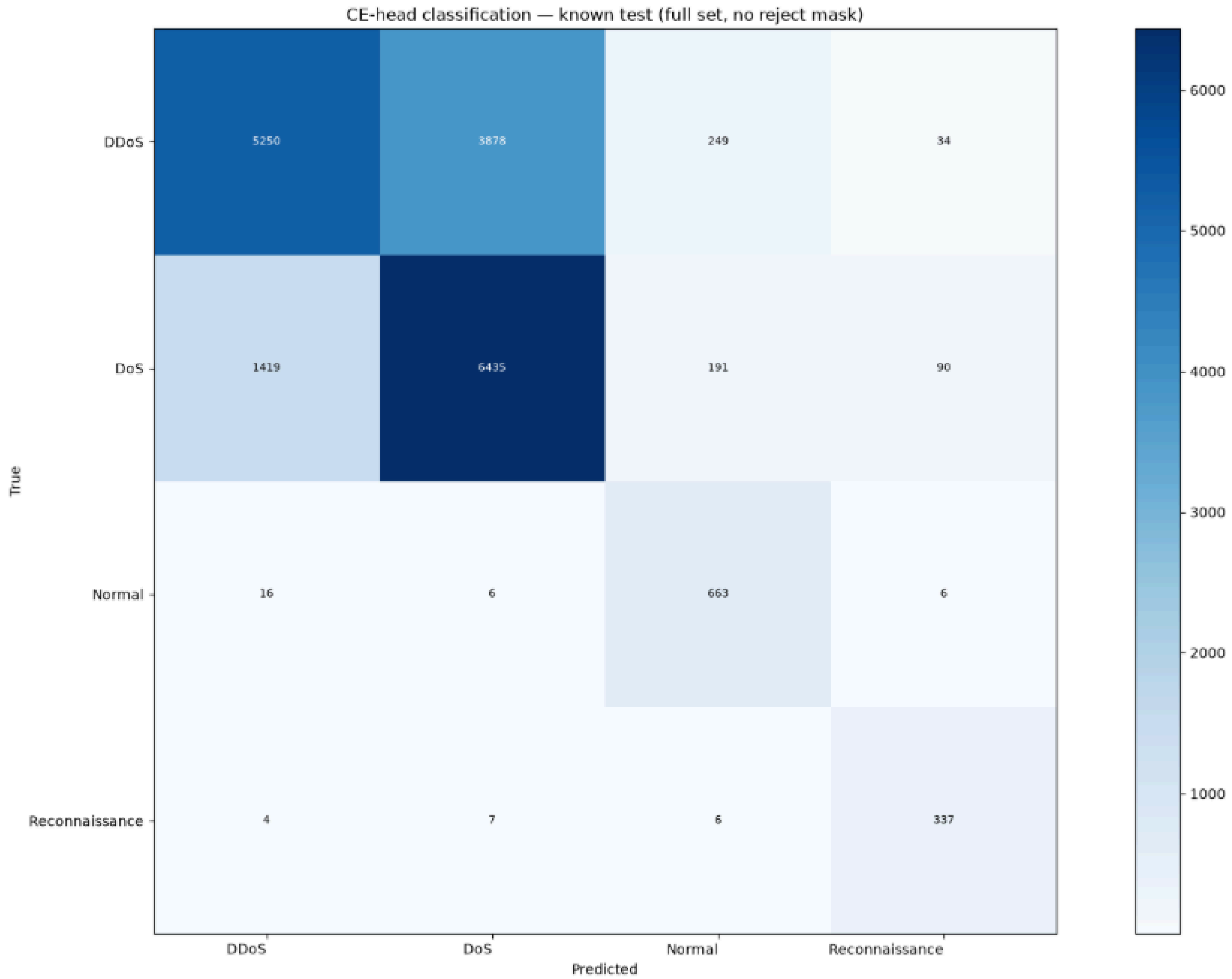
Observation: more train samples lift CE full-test accuracy by about +3 points vs the previous 3000-coreset run. Normal and Reconnaissance look strong on this matrix (high recall); the main residual error is still DDoS ↔ DoS swap.

```
[test CE] accuracy (full set, no reject mask): 0.6823
[test CE] macro-F1 (full set): 0.7267
[test CE] n=18591

[test CE] classification report (full set):
                precision    recall  f1-score   support

          DDoS       0.78      0.56      0.65      9411
           DoS       0.62      0.79      0.70      8135
        Normal       0.60      0.96      0.74       691
Reconnaissance       0.72      0.95      0.82       354

      accuracy                           0.68     18591
     macro avg       0.68      0.82      0.73     18591
  weighted avg       0.71      0.68      0.68     18591
```



Current working choice:

Knob	Choice
Classes	4 (DDoS, DoS, Normal, Reconnaissance)
Train data	full FROZEN known train (no per-class cap)
LR	cosine schedule, start 0.01
Notebooks / artefacts	`final-bot-iot-maqt` / `4class-all-lr0.01auto-*`

Decision trail (short)

```
4c / 3k / lr=0.05
  $\rightarrow$ Normal weak, DDoS↔DoS confused
3c (drop DoS) / 3k / lr=0.05
  $\rightarrow$ Normal still not fixed $\rightarrow$ back to 4c
```

```
4c / 20k / lr=0.05
  $\rightarrow$ accuracy up; DDoS↔DoS / Normal still hard
4c / 10k / lr=0.0005
  $\rightarrow$ ≈ 3k result; not worth the extra data for this goal
4c / 3k / cosine 0.03 vs 0.01
  $\rightarrow$ similar accepted metrics; prefer 0.01 (smoother)
previous latest: 4c / 3k / cosine start 0.01 (~65% CE full-test vs arjun ~64%)
current latest: 4c / full train / cosine start 0.01
  $\rightarrow$ ~68% CE full-test (no reject); Normal/Recon strong; DDoS↔DoS still open
```